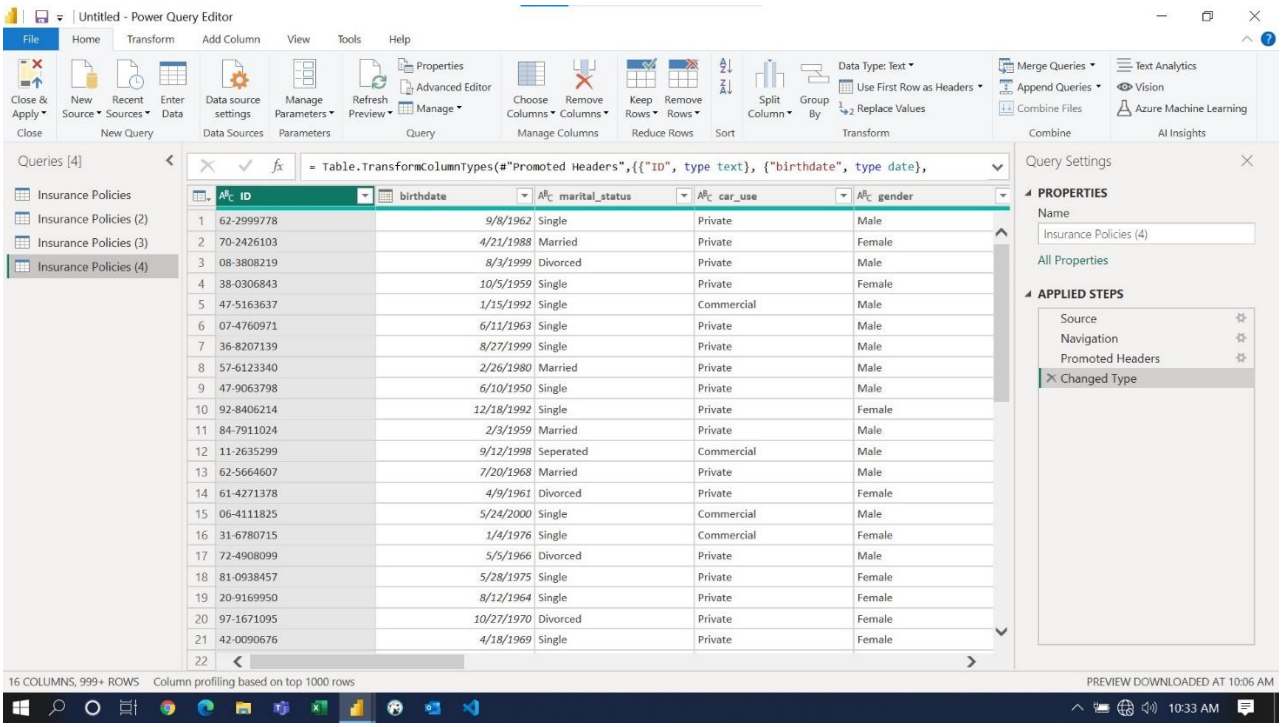
DATA SOURCE

The provider of the Data is **Onyx Data Limited** (Company No. 11795932) whose registered office address is 3rd Floor Buckingham House, Buckingham Street, Aylesbury, Buckinghamshire, United Kingdom, HP20 2LA ("Promoter"). The data used is of an insurance company giving out its data for insights and recommendation on how to maximize profit and satisfy its policy holders

DATA PROCESSING

CLEANING: Transformation of data is firstly done in the power query engine of power bi. the data was cleaned by trimming, when necessary, removal of duplicated customers id from the table, and splitting and change of data types



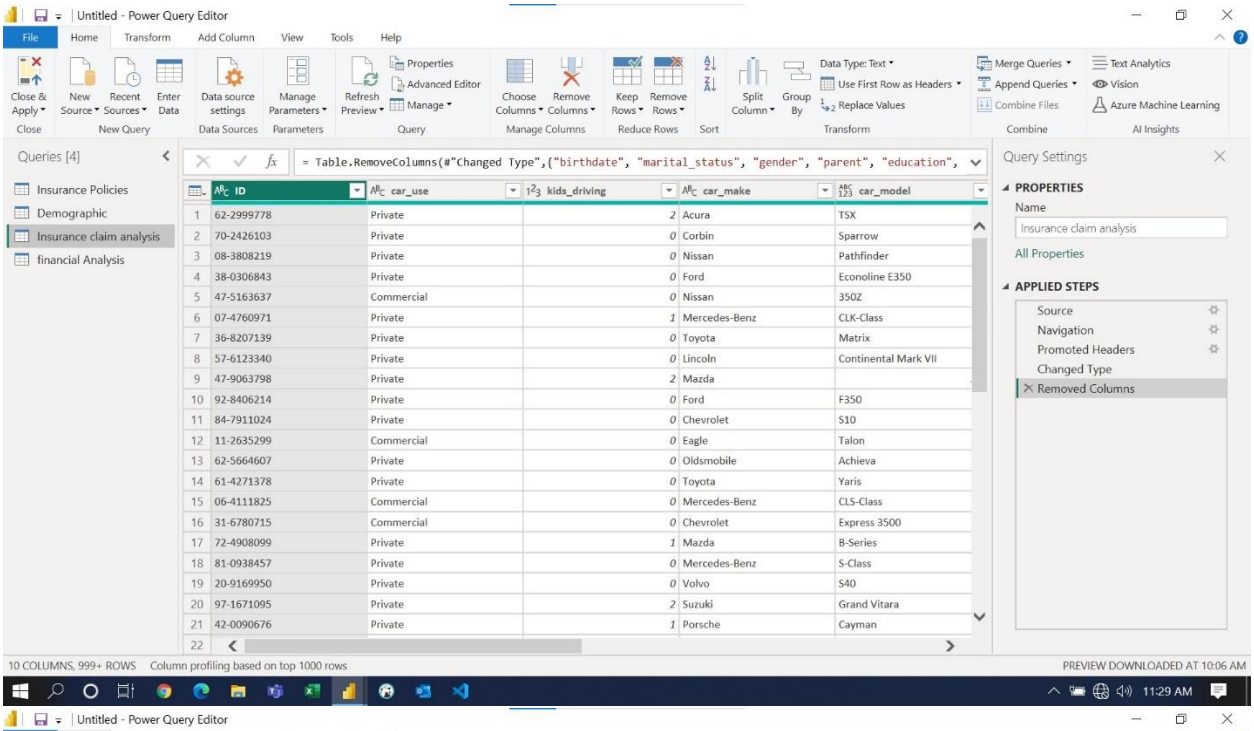
	ID	birthdate	marital_status	car_use	gender
1	62-2999778	9/8/1962	Single	Private	Male
2	70-2426103	4/21/1988	Married	Private	Female
3	08-3808219	8/3/1999	Divorced	Private	Male
4	38-0306843	10/5/1959	Single	Private	Female
5	47-5163637	1/15/1992	Single	Commercial	Male
6	07-4760971	6/11/1963	Single	Private	Male
7	36-8207139	8/27/1999	Single	Private	Male
8	57-6123340	2/26/1980	Married	Private	Male
9	47-9063798	6/10/1950	Single	Private	Male
10	92-8406214	12/18/1992	Single	Private	Female
11	84-7911024	2/3/1959	Married	Private	Male
12	11-2635299	9/12/1998	Seperated	Commercial	Male
13	62-5664607	7/20/1968	Married	Private	Male
14	61-4271378	4/9/1961	Divorced	Private	Female
15	06-4111825	5/24/2000	Single	Commercial	Male
16	31-6780715	1/4/1976	Single	Commercial	Female
17	72-4908099	5/5/1966	Divorced	Private	Male
18	81-0938457	5/28/1975	Single	Private	Female
19	20-9169950	8/12/1964	Single	Private	Female
20	97-1671095	10/27/1970	Divorced	Private	Female
21	42-0090676	4/18/1969	Single	Private	Female
22					

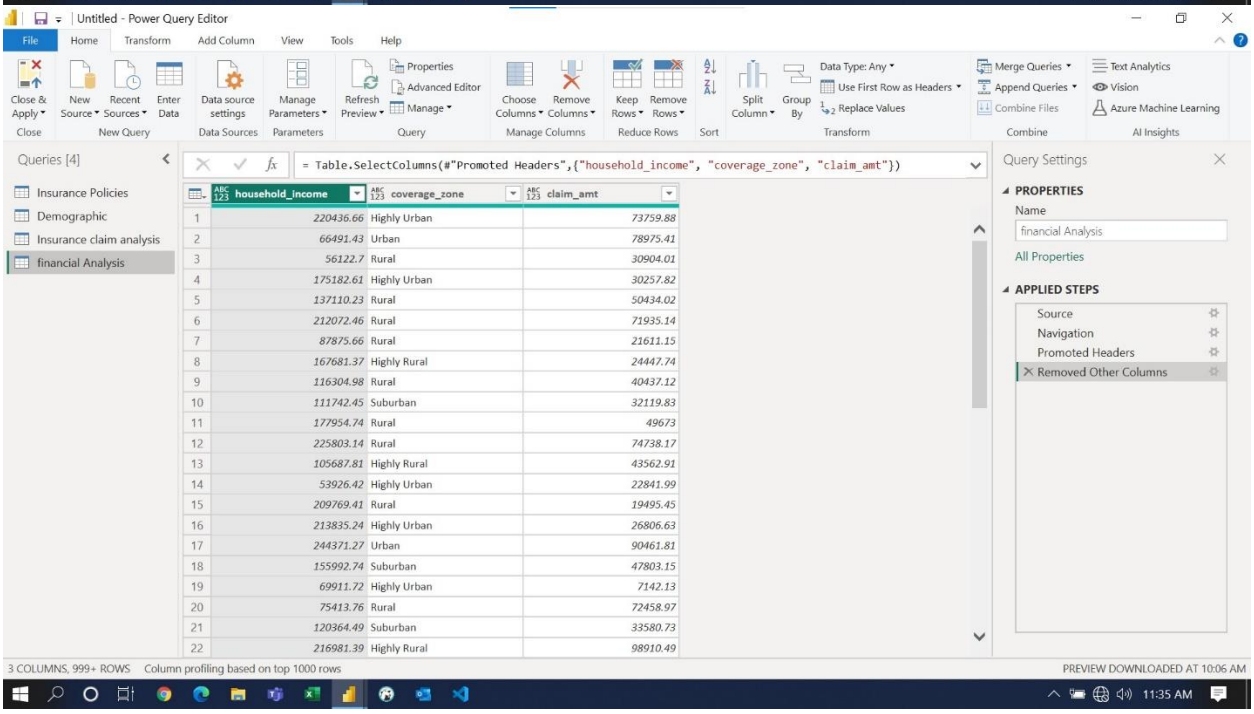
SPLITING INTO TABLES: Creation of different tables from the dataset. Tables created are the Demographic Table, insurance Claim analysis table and the financial analysis table. The reason for this is to have full focus on separate areas of the big data in order to get the best analysis possible out of it. The Tables are separated with the customer's ID as the primary key.

DEMOGRAPHIC DATA: Id, Birthdate, Marital Status, Gender, Parent, Education, Coverage Zone.

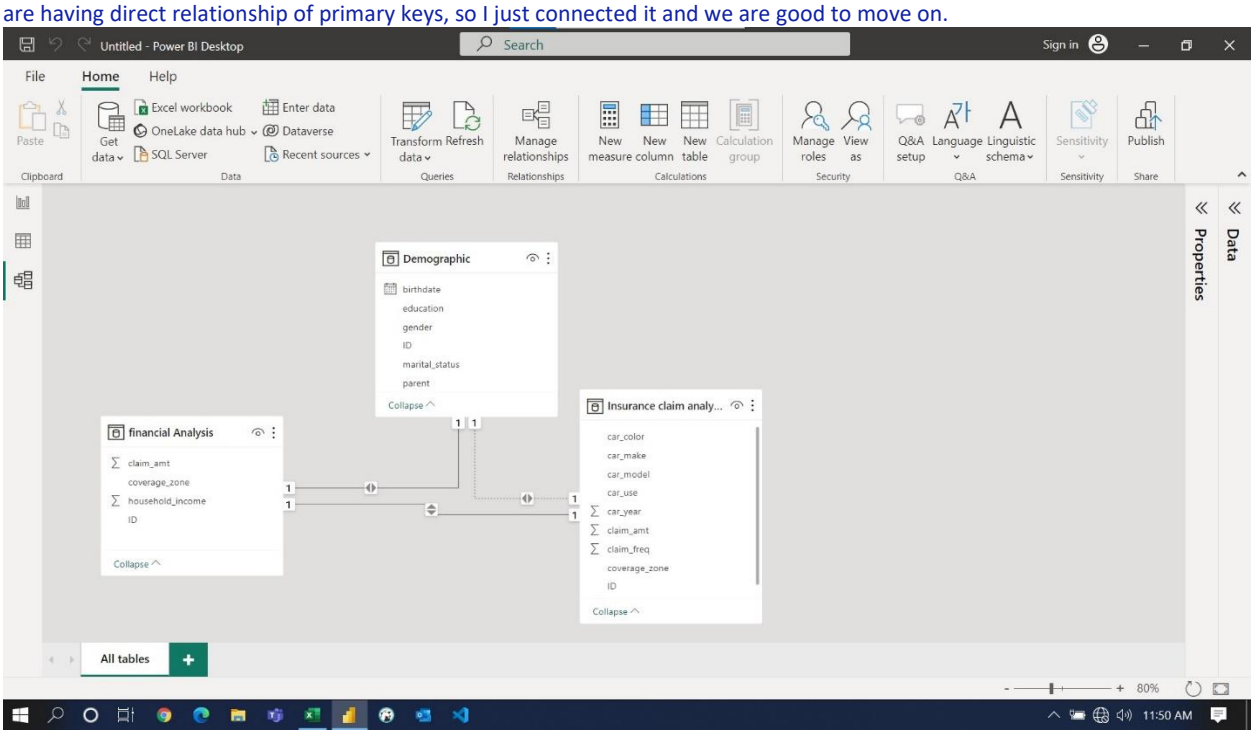
INSURANCE CLAIM: Id, Car_Use, Kid_Driving, Car_Make, Car_Model, Car_Color, Car_Year, Claim_Freq, Coverage Zone, Claim_Amt

FINANCIAL ANALYSIS: Id, Household_Income, Coverage_Zone, Claim_Amt


Untitled - Power Query Editor
File Home Transform Add Column View Tools Help
Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Text Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning
Close New Query Data Sources Parameters Query Manage Columns Reduce Rows Sort Transform Combine AI Insights
Queries [4]
Insurance Policies
Demographic
Insurance claim analysis
financial Analysis
= Table.RemoveColumns(#"Changed Type",{"birthdate", "marital_status", "gender", "parent", "education",
ID car_use kids_driving car_make car_model
1 62-2999778 Private 2 Acura TSX
2 70-2426103 Private 0 Corbin Sparrow
3 08-3808219 Private 0 Nissan Pathfinder
4 38-0306843 Private 0 Ford Econoline E350
5 47-5163637 Commercial 0 Nissan 350Z
6 07-4760971 Private 1 Mercedes-Benz CLK-Class
7 36-8207139 Private 0 Toyota Matrix
8 57-6123340 Private 0 Lincoln Continental Mark VII
9 47-9063798 Private 2 Mazda
10 92-8406214 Private 0 Ford F350
11 84-7911024 Private 0 Chevrolet S10
12 11-2635299 Commercial 0 Eagle Talon
13 62-5664607 Private 0 Oldsmobile Achieva
14 61-4271378 Private 0 Toyota Yaris
15 06-4111825 Commercial 0 Mercedes-Benz CLS-Class
16 31-6780715 Commercial 0 Chevrolet Express 3500
17 72-4908099 Private 1 Mazda B-Series
18 81-0938457 Private 0 Mercedes-Benz S-Class
19 20-9169950 Private 0 Volvo S40
20 97-1671095 Private 2 Suzuki Grand Vitara
21 42-0090676 Private 1 Porsche Cayman
22
Query Settings
PROPERTIES
Name
Insurance claim analysis
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
Changed Type
Removed Columns
10 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 10:06 AM
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Untitled - Power Query Editor
File Home Transform Add Column View Tools Help
Close & Apply New Source Recent Sources Enter Data Data source settings Manage Parameters Refresh Preview Properties Advanced Editor Manage Choose Columns Remove Columns Keep Rows Remove Rows Split Column Group By Data Type: Any Use First Row as Headers Replace Values Merge Queries Append Queries Combine Files Text Analytics Vision Azure Machine Learning
Close New Query Data Sources Parameters Query Manage Columns Reduce Rows Sort Transform Combine AI Insights
Queries [4]
Insurance Policies
Demographic
Insurance claim analysis
financial Analysis
= Table.SelectColumns(#"Promoted Headers",{"household_income", "coverage_zone", "claim_amt"})
household_income coverage_zone claim_amt
1 220436.66 Highly Urban 73759.88
2 66491.43 Urban 78975.41
3 56122.7 Rural 30904.01
4 175182.61 Highly Urban 30257.82
5 137110.23 Rural 50434.02
6 212072.46 Rural 71935.14
7 87875.66 Rural 21611.15
8 167681.37 Highly Rural 24447.74
9 116304.98 Rural 40437.12
10 111742.45 Suburban 32119.83
11 177954.74 Rural 49673
12 225803.14 Rural 74738.17
13 105687.81 Highly Rural 43562.91
14 53926.42 Highly Urban 22841.99
15 209769.41 Rural 19495.45
16 213835.24 Highly Urban 26806.63
17 244371.27 Urban 90461.81
18 155992.74 Suburban 47803.15
19 69911.72 Highly Urban 7142.13
20 75413.76 Rural 72458.97
21 120364.49 Suburban 33580.73
22 216981.39 Highly Rural 98910.49
Query Settings
PROPERTIES
Name
financial Analysis
All Properties
APPLIED STEPS
Source
Navigation
Promoted Headers
Removed Other Columns
3 COLUMNS, 999+ ROWS Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 10:06 AM
11:35 AM

MODELLING: This modelling aspect of the project is very simple and straight forward, since the tables are not much and are having direct relationship of primary keys, so I just connected it and we are good to move on.



CALCULATIONS AND MEASURES ON EACH TABLE

In this section, I jotted out all what's needed to be generated in terms of measure and calculated columns so as to make my work more precise, focused and timely managed. Though a few of them were also later added along during the report processing.

DEMOGRAPHICS*:* Age(Substracting The Birthdate From Current Year), Age Bracket(0-25{Youth},26-45{Adult},46-65{Young_Old},>66{Old}), Average_Age, B.Sc Educational Level, Count Of Parent Status.

FINANCIAL CLAIM: Average Claim Amount, Average Income, Claim Frequency(Related From Another Table(Insurance)),Total Policy Holders, %Paid_Cars, Total Claim Amount, Count Of Cars,Average Household Income And Average Claim_Amount

INSURANCE CLAIM*:* Average Car_Age, Average, Claims,Car_Age, Car_Year _Group,Claim Amount Paid And Claim Amount Unpaid, Claim Count Unpaid, Claim Profit, Max And Min Claim Amount, Total Profit.

FORD*:* Top5(Ford Brand With Highest Claimed Amount)

MEASURES AND CALCULATED COLUMNS USED AND THEIR FORMULAR.

CALCULATED COLUMNS:

```
#Age = -DATEDIFF(TODAY(),[birthdate],YEAR)
#Age_bracket = IF([Age]<26,"youth",if([Age]<46,"Adult",if([Age]<66,"young_old","old")))
#claim_Frequency = RELATED('Insurance claim analysis'[claim_freq])
# Car_age = YEAR(TODAY())-[car_year]
# Car_year_group =
IF([car_year]<=1920,"20s",IF([car_year]<=1940,"40s",IF([car_year]<=1960,"60s",IF([car_year]<=1980,"80s",IF([car_year]<=2000,
"2000s",IF([car_year]<=2020,"2020s","Latest"))))))
# claim_amt_paidc = [claim_amt]*[claim_freq]
(MEASURES) claim_amt_unpaid = CALCULATE(SUM([claim_amt]), 'Insurance claim analysis'[claim_freq]=0)
```

CALCULATED TABLE:

```
#ford =
var _filteredtable= FILTER('Insurance claim analysis',[car_make]="ford")
Var _grouping=
SUMMARIZE(
   _filteredtable,
   [car_model],
   "amountpaid", SUM('Insurance claim analysis'[claim_amt_paidc])
)
return
_grouping
```

MEASURES:

```
#Avg_age = AVERAGE(Demographic[Age])

#B.sc_levels% = CALCULATE(
var _Total=COUNTA(Demographic[education])
 var _Bscno= CALCULATE(COUNT(Demographic[education]),Demographic[education]="Bachelors")
 return
 DIVIDE(_Bscno,_Total),ALL(Demographic[education]))


#count_parent = DIVIDE(CALCULATE(COUNT('Demographic'[parent]), FILTER(Demographic,[parent]="yes")), COUNT([parent]))


# Avg_claims_amt = AVERAGE('financial Analysis'[claim_amt])

#Avg_H.income = AVERAGE('financial Analysis'[household_income])


#TOP5 = CALCULATE(SUM(ford[amountpaid]),

          TOPN(5,'ford',[amountpaid],DESC)
      )

#Avg_car_age = AVERAGE('Insurance claim analysis'[Car_age])


#Avg_claims = AVERAGE('financial Analysis'[claim_amt])

#claim_count_unpaid = CALCULATE(COUNT([claim_amt]), 'Insurance claim analysis'[claim_freq]=0)
```

```
#claim_profit = IF([claim_amt_paidc]=0,[claim_amt],-[claim_amt_paidc])

#Max_claim_amt = MAX('Insurance claim analysis'[claim_amt])

#Min_claim_amt = MIN('Insurance claim analysis'[claim_amt])

#total_profit = SUM('Insurance claim analysis'[claim_profit])
#%paid_cars = DIVIDE(

            CALCULATE(
                COUNT('Insurance claim analysis'[claim_amt_paidc]),
                FILTER('Insurance claim analysis',[claim_amt_paidc]>0)
               ),
            count('Insurance claim analysis'[claim_amt_paidc]
            )
          )
#Total claim amount for first time = SUM('financial Analysis'[claim_amt])

#Count_of_cars = DISTINCTCOUNT('Insurance claim analysis'[car_make])

#AVG_car_age of paid_amt = AVERAGEX(FILTER('Insurance claim analysis','Insurance claim
analysis'[claim_amt_paidc]>0),'Insurance claim analysis'[Car_age])`

#Total policy holder = COUNT(Demographic[ID])
```

DESIGN

Canvas background: #ebecee
Overlay background 1: #fffff
Overlay background 2: #e8f2fc
Color mixtures for columns, bars and lines: #1c1f33, #e86800, #9b7874, #a56749, #cc9848.

REPORT VIEW

Find below the report charts…

DATA *DATASET FILE*, *VISUALIZATION FILE*, *VISUALIZATION VIEW*


KEY INSIGHT:
• An approximate of 38,000 plicy holders has insured their cars of 78 different maker, 1009 different model and the least estimation to be paid for the first time if all cars got damaged is $1.88bn
• There is no significant difference in the ratio of gender differences of policy holders(male 49.9%, Female 50.1%)
• B.Sc holders constitute half of the whole population
• Only 44% of the population are parents
• Ford motors is insured most , Mustang model is insured most and got paid most of the claims
• Average age of cars with most claims is 23year
• Private cars in urban zone has more paid claims than others of about 83.71%
OLANIYI MICHEAL OLUWATOSIN
car_use
All
car_model
All
Car Manufac...
All
INSURING SUCCESS
Insights For Car Insurance Claims
Cover Page
Demographics
Insurance claim
Financial Analysis
Recomendations
47
Average Age
$147.25K
Avg H.hold Income
50%
B.sc level of claimers
44%
parental policy holders
23
Average Car Age
$399M
Total profit
$100K
Max claim Amount
$0.04
Min claim Amount
23
Average age of car paid of claims
28%
Percentage of paid cars
78
Distinct car manufacturer
$1.88bn
Total claim amount for first time
38K
Total policy holders

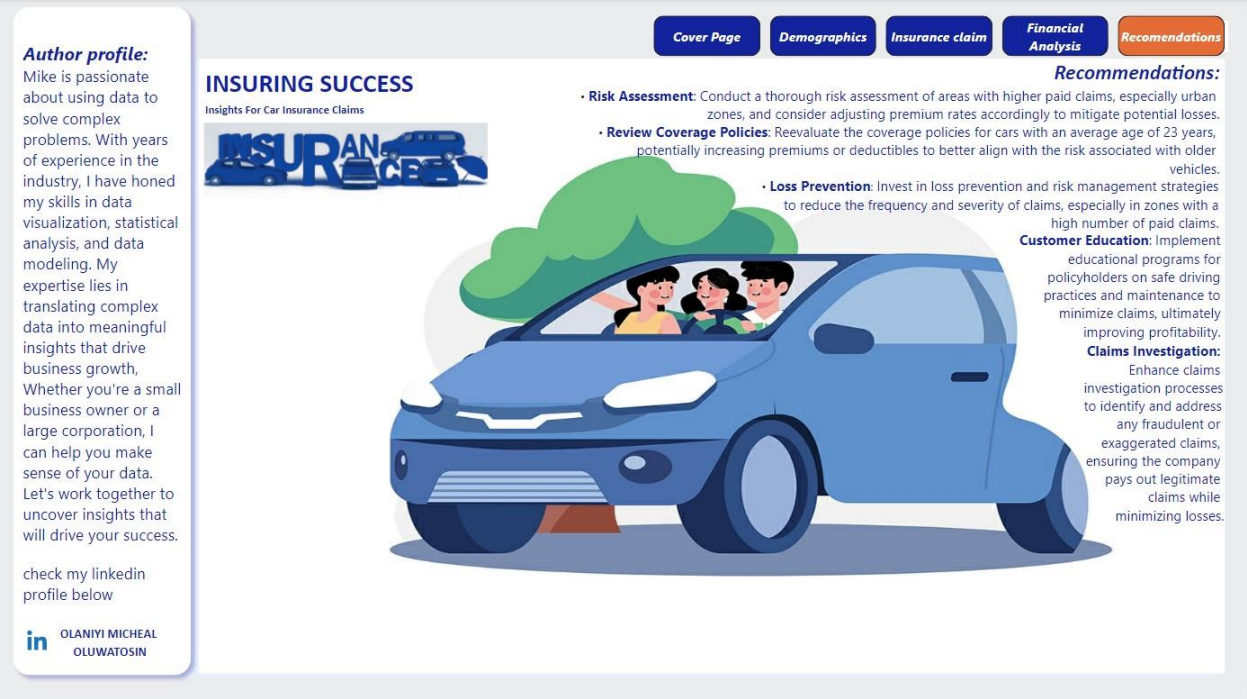

DEMOGRAPHICS:
• Average age of their place holders is 47years old and
• The level of bachelors degree holder in this company as a placeholder is exactly the same as the summation of the rest of the placeholders educational count.
• The average income of these set of people is $147,000
• Most of the placeholders are not parents, most of them are single and they occupy only 44% of the totality.
• There is no gender bias in the rate of insurance for either female or male as they both have the fair share of their portion in the distribution (50/50).
• Lastly the single bachelors have more good income compare to others of the different marital status or education
OLANIYI MICHEAL OLUWATOSIN
car_use
All
car_model
All
Car Manufac...
All
Cover Page
Demographics
Insurance claim
Financial Analysis
Recomendations
47
Average Age
$147.25K
Avg H.hold Income
50%
B.sc level of claimers
44%
parental policy holders
young_old 14.2K
Adult 14.1K
old 5.8K
youth 3.4K
Customer by Age Range
Bachel... 19K
High S... 10K
Masters 6K
PhD 3K
Customer by Education
Female Male
49.9%
50.1%
Customer by Gender
parent No Yes
10K
5K
0K
Single
Married
Divorced
Seperated
Customer's count of Marital status in different parent status
marital_status Divorced Married Seperated Single
1.0bn
0.5bn
0.0bn
Bachelors
High School
Masters
PhD
Customer's income by Education and marital status





Customer ID	gender	Age_bracket	Sum of Age	marital_status	education	parent	car_Manufacturer	car_model	Claim_freq	Claim_amt_paidc
84-9191366	Male	young_old	51	Married	Bachelors	No	Buick	Skylark	4	399921
13-8497885	Male	young_old	53	Married	PhD	No	Geo	Metro	4	399699
13-0252427	Female	young_old	57	Single	Bachelors	No	Suzuki	Verona	4	399476
89-9634536	Female	youth	22	Divorced	Bachelors	No	Lincoln	Continental	4	399476
57-9717346	Female	young_old	47	Single	High School	Yes	Chevrolet	1500	4	399409
49-6684343	Female	old	71	Single	Bachelors	Yes	Dodge	Caravan	4	399118
86-2233627	Male	young_old	53	Single	Bachelors	No	Volvo	960	4	399040
63-8324121	Male	Adult	41	Single	High School	No	Lexus	GS	4	398860
70-5443005	Male	young_old	49	Married	Bachelors	No	Chevrolet	Monte Carlo	4	398586
35-9676144	Female	Adult	35	Single	PhD	No	Ford	F150	4	398500

Top 10 Max claimed insured amount Customers' Details



DATA STORYLINE

DEMOGRAPHICS:

from the data gotten from this insurance company, the average age of their place holders is 47years old and that means according to our age range grouping, all most of them are categorically and adult. The level of bachelors degree holder in this company as a placeholder is exactly the same as the summation of the rest of the placeholders educational count. this means that most of the people who has the idea to go for an insurance are mostly BSc holders and the average income of these set of people is $147,000

Another aspect to this data is that most of the placeholders are not parents, most of them are single and they occupy only 44% of the totality.

Many of their customer's are single, Married, Divorced and Separated in that descending order of population so also in every aspect of the marital stages that the rate of being a parent is lesser .

In addition, there is no gender bias in the rate of insurance for either female or male as they both have the fair share of their portion in the distribution (50/50).

Still on the demographic aspect, its good to know that the single bachelors have more good income compare to others of the different marital status or education

INSURANCE CLAIMS:

The Average age of their place holders is 47years old and the totality of the insurance company generates over the years from placeholders who have insured but has not come back with and proof to claim is $399m. the maximum amount of claim that has ever been agreed to be paid to a customer is $100,000 while the least is $0.04.

this company has really been dormant and stagnant over the years with little or no productivity until the early 80's when the company started generating revenues.

more than 90% of the profit made by the company is generated is in the last 40 years.

The car maker with claimed amount and frequency in claims is "FORD" and the top 5 ford products with such high insurance statistic are "Mustang, Ranger, Thunderbird, Taurus and f150"

Lastly the amount of fee claimed by gender is of the same ratio ($0.48billion each)

Financial analysis:

Analysis has shown that most of the amount claimed and paid out across various coverage zones is dominated by the private owners compared to the public ones
Most of the people between the age of 46 to 65 are the ones who insured most without being paid
the descending hierarchy of top 5 manufacturer with highest claimed amount is "ford, cheverolet, dodge, gmc, toyota"
Age bracket "young old" constitute the highest percentage of top10 customers with highest claim amount.
78 different cars were insured but only 28% of it got paid for insurance.
the total claim amount for every cars to be paid on their first time is $1.88bn

KEY INSIGHT:

- *An approximate of **38,000** policy holders has insured their cars of **78** different maker, **1009** different model and the least estimation to be paid for the first time if all cars got damaged is **$1.88bn***
- *There is no significant difference in the ratio of gender differences of policy holders(male **49.9%,** Female **50.1%**)*
- *B.Sc holders constitute half of the whole population*
- *Only **44%** of the population are parents*
- *Ford motors is insured most , Mustang model is insured most and got paid most of the claims*
- *Average age of cars with most claims is **23year***
- *Private cars in urban zone has more paid claims than others of about **83.71%***

RECOMMENDATIONS:

- *Risk Assessment: Conduct a thorough risk assessment of areas with higher paid claims, especially urban zones, and consider adjusting premium rates accordingly to mitigate potential losses.*
- *Review Coverage Policies: Reevaluate the coverage policies for cars with an average age of 23 years, potentially increasing premiums or deductibles to better align with the risk associated with older vehicles.*
- *Loss Prevention: Invest in loss prevention and risk management strategies to reduce the frequency and severity of claims, especially in zones with a high number of paid claims.*
- *Customer Education: Implement educational programs for policyholders on safe driving practices and maintenance to minimize claims, ultimately improving profitability.*
- *Claims Investigation: Enhance claims investigation processes to identify and address any fraudulent or exaggerated claims, ensuring the company pays out legitimate claims while minimizing losses.*

About the Analyst:

Olaniyi Micheal Oluwatosin

Mike is passionate about using data to solve complex problems. With years of experience in the industry, I have honed my skills in data visualization, statistical analysis, and data modeling. My expertise lies in translating complex data into meaningful insights that drive business growth.
Whether you're a small business owner or a large corporation, I can help you make sense of your data. Let's work together to uncover insights that will drive your success.
Contact me to learn more about how I can help you achieve your data goals